Reuben Abraham T. Georgi

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Robotics Engineer with 4 years of experience and a Master's degree in Control and Autonomy, specializing in perception, planning, and control for autonomous systems. Experienced in building navigation and manipulation pipelines, integrating classical methods with modern learning approaches. Focused on developing safe and reliable autonomous systems for real-world deployment.

EDUCATION

Purdue University — West Lafayette, USA
Master of Science in Aeronautics and Astronautics (MSAA) — May 2021

Punjab Engineering College — Chandigarh, India
Bachelor of Technology (B.Tech) in Aerospace Engineering — October 2019

WORK EXPERIENCE

Simelabs (Astek Group) — Kochi/Bangalore, India
Robotics Systems Engineer — February 2022 - Present

AI-Driven Robotic Inventory Station (At **West Pharmaceutical Services**)

- Developed a vision-guided robotic inventory system enabling autonomous item deposit and retrieval for site operators
- Integrated a Vision-Language Model (VLM) to support natural language interaction for object retrieval and general query handling
- Designed a perception pipeline using SAM (Segment Anything) and homography to estimate object coordinates for precise robotic alignment and manipulation using a single robot-mounted camera
- Optimized system performance to achieve average response times of under 5 seconds across retrieval and query tasks

R&D Robotic Device Assembly Line - Selected Features (At **West Pharmaceutical Services**)

- Created a component pose estimation pipeline for autonomous clearing using a RealSense RGB-D camera and ABB YuMi; leveraged Open3D (RANSAC, ICP), YOLO detection, and SAM segmentation to achieve 5-10 mm pose accuracy under fixed camera and working distance constraints
- Developed ROS2 nodes and a Docker image using MoveIt 2 to ensure collision-free trajectories during line restarts, reducing manual intervention and programming effort
- Implemented real-time remote visualization of assembly operations in RobotStudio by integrating robot and PLC data

Autonomous Mobile Robot system with Vision-Based Decision Making

- Implemented ROS 2 perception nodes for ArUco markers and environmental cues (lights, signs, timed gates) using OpenCV and Keras models for mobile robot (TurtleBot3)
- Developed a high-level behavior planner using Nav2 and custom ROS 2 nodes to execute user-defined waypoints and autonomously navigate based on vision-derived cues
- Mitigated camera latency and odometry-related localization issues, and delivered a system usable via a custom PyQt testing GUI or an external Android application

Automation Simulations and System Validation (At **West Pharmaceutical Services**)

- Simulated automation systems in RoboDK and RobotStudio for inspection, material handling, and packaging applications to evaluate feasibility, validate layouts, and optimize cycle time and costs

INTERNSHIPS

- **AbbVie Inc.** (Chicago, USA | 2020) **-** CFD simulations with ANSYS and M-Star for research study to characterize hydrodynamics of USP dissolution apparatus
- **Hindustan Aeronautics Limited (**Bangalore, India | 2018) **-** Helicopter rotor mathematical modelling and numerical simulation to estimate performance
- **Khalifa University (SPACE Lab**) (Abu Dhabi, UAE | 2017) **-** Line follower robot and CubeSat setup

SELECTED PROJECTS

UAV Simulation and Motion Planning Framework ***(Link)***

- Developing a modular and extensible Python tool for simulating quadcopter motion and experimenting with different motion planning and control techniques
- Currently implemented features:
  - Simulation: Set actuator limits, actuator delays, waypoints, wind disturbances (constant or Gaussian), obstacles (spherical)
  - Planning: Dijkstra, A*, RRT*, iLQR (Iterative Linear Quadratic Regulator)
  - Control: Cascade PID, LQR (using JAX for autodiff), linear MPC (using CasADi for optimization)
  - Visualization: Plots of state variable evolution with time, animation of quadcopter flight

Autonomous Mobile Manipulator Simulation ***(Link)***

- Designed and implemented a ROS 2 mobile manipulator system integrating Nav2, MoveIt 2, and Gazebo to enable autonomous navigation and manipulation
- Utilized BehaviorTree.CPP to react to dynamically spawned objects and orchestrate task-level actions

3D SLAM ***(Link)***

- Implemented the extended and unscented Kalman filter (EKF, UKF) in Python for simultaneous localization and mapping in 3 dimensions
- Simulated scenarios with known correspondences and unknown number of landmarks and introduced uncertainty by adding noise to odometry and sensor measurements

Quadcopter Design using Genetic Algorithms ***(Link)***

- Formulated a multi-objective problem to minimize cost and weight of a quadcopter subject to performance and integration constraints
- Developed MATLAB app for users to provide data and to visualize results, including the Pareto frontier which displays Pareto optimal solutions.

TECHNICAL SKILLS

Languages: **Python**, **C++, MATLAB**

Packages/Libraries: **Nav2 (ROS 2), MoveIt 2 (ROS 2), Gazebo, OpenCV, JAX, BehaviorTree.CPP**

Other software: **ABB RobotStudio/RAPID**, **RoboDK**, **Simulink, ANSYS**

PUBLICATIONS

Kubinski, A.M., Sosa, R.D., Shivkumar, G., **Georgi, R.**, et al., ***Predictive dissolution modeling across USP apparatuses I, II, and III***, Journal of Pharmaceutical Sciences, 2025. https://doi.org/10.1016/j.xphs.2025.103765

Kubinski, A.M., Shivkumar, G., **Georgi, R.A.**, et al., ***Predictive Drug Release Modeling Across Dissolution Apparatuses I and II***, Journal of Pharmaceutical Sciences, 2023. https://doi.org/10.1016/j.xphs.2022.10.027